Yuyang Dong (David Dong)

Professional Experience

The University of Osaka — Invited Associate Professor May 2026 – Present

SB Intuitions (SoftBank Group) Sep 2025 – Jun 2026
Chief Research Scientist; LLM Team Tech Lead

- **Sarashina — Model Development Lead.** Led end-to-end development of Japanese LLMs from scratch. Sarashina3 mini / nano combined 30T-token pretraining, multi-stage reinforcement learning, and tool-using agent capabilities. **[Official technical blog]**

NEC Corporation Apr 2019 – Sep 2025
Generative AI Center / Data Science Research Laboratories
Principal Researcher (Apr 2024 – Sep 2025); Special Researcher (Apr 2021 – Mar 2024);
Researcher (Apr 2019 – Mar 2021).

- **cotomi / cotomi-v2 — Core Team Member.** Contributed to Japanese LLM development across data curation, model training, and evaluation. **[NEC interview]**
- **Chart & Document Understanding.** Developed VLM-based understanding of charts, tables, and diagrams, preserving structural context for document transcription and multimodal RAG. **[Technology announcement]**
- **Data Discovery & Preprocessing.** Developed DeepJoin for table discovery and co-developed Jellyfish for LLM-based data preprocessing.

Education

Ph.D. in Computer Science, University of Tsukuba 2016 – 2019

Master of Engineering in Computer Science, University of Tsukuba 2014 – 2016

Bachelor's degree in Computer Science 2013
China University of Mining & Technology, Beijing

Selected Publications

*Equal contribution.

1. **Adaptive Layer Selection for Layer-Wise Token Pruning in LLM Inference**
   R. Taniguchi, **Yuyang Dong**, M. Onizuka, C. Xiao. Findings of ACL, 2026, pp. 967–986.
2. **SCAN: Semantic Document Layout Analysis for Textual and Visual Retrieval-Augmented Generation**
   N. Ueda*, **Yuyang Dong***, K. Boros, D. Ito, T. Sera, M. Oyamada. Findings of EACL, 2026, pp. 1618–1637.
3. **Jellyfish: Instruction-Tuning Local Large Language Models for Data Preprocessing**
   H. Zhang*, **Yuyang Dong***, C. Xiao, M. Oyamada. EMNLP, 2024, pp. 8754–8782.
4. **DeepJoin: Joinable Table Discovery with Pre-trained Language Models**
   Yuyang Dong, C. Xiao, T. Nozawa, M. Enomoto, M. Oyamada. Proceedings of the VLDB Endowment, 16(10), 2023, pp. 2458–2470.
5. **Table Enrichment System for Machine Learning**
   Yuyang Dong, M. Oyamada. SIGIR, 2022, pp. 3267–3271; system demonstration.
6. **Efficient Joinable Table Discovery in Data Lakes: A High-Dimensional Similarity-Based Approach**
   Yuyang Dong, K. Takeoka, C. Xiao, M. Oyamada. ICDE, 2021, pp. 456–467.
7. **Aggregate Reverse Rank Queries**
   Yuyang Dong, H. Chen, K. Furuse, H. Kitagawa. DEXA, 2016, pp. 87–101; **Best Paper Award**.
8. **Learning Deliberately, Acting Intuitively: Unlocking Test-Time Reasoning in Multimodal LLMs**
   Y. Yu, **Yuyang Dong**, M. Oyamada. arXiv:2507.06999, 2025; preprint.

Honors & Awards

DBSJ Kambayashi Young Researcher Award 2022
President's Award, University of Tsukuba 2019
Best Paper Award, DEXA — *Aggregate Reverse Rank Queries* 2016